

-- Fix missing or zero quantities

Phase 1: Data Cleaning SQL:

```
UPDATE orders_raw
SET quantity = 1
WHERE quantity IS NULL OR quantity = 0;


SELECT COUNT(*) as missing_customers
FROM orders_raw
WHERE customer_id IS NULL;


-- First, see which rows have the wrong date format
SELECT order_id, order_date
FROM orders_raw
WHERE order_date LIKE '%/%'
LIMIT 20;


-- Convert MM/DD/YYYY dates to YYYY-MM-DD
UPDATE orders_raw
SET order_date =
    substr(order_date, 7, 4) || '-' ||
    substr(order_date, 1, 2) || '-' ||
    substr(order_date, 4, 2)
WHERE order_date LIKE '%/%';
```

```sql
SELECT order_id, order_date
FROM orders_raw
WHERE order_date LIKE '%/%';

SELECT MIN(order_date) as earliest, MAX(order_date) as latest
FROM orders_raw;

SELECT order_id, COUNT(*) as times_appeared
FROM orders_raw
GROUP BY order_id
HAVING COUNT(*) > 1
ORDER BY times_appeared DESC;

-- Create a clean version of the table without duplicates
CREATE TABLE orders_clean AS
SELECT DISTINCT *
FROM orders_raw;

SELECT
   (SELECT COUNT(*) FROM orders_raw) as original_rows,
   (SELECT COUNT(*) FROM orders_clean) as cleaned_rows;
```

Final Cleaning Polish SQL:

```sql
UPDATE orders_clean
SET quantity = CAST(quantity AS INTEGER);

SELECT
   COUNT(*) as total_orders,
   COUNT(DISTINCT order_id) as unique_orders,
   COUNT(DISTINCT customer_id) as unique_customers,
   SUM(CASE WHEN customer_id IS NULL THEN 1 ELSE 0 END) as missing_customer_ids,
   MIN(order_date) as earliest_date,
   MAX(order_date) as latest_date,
   SUM(line_total) as total_revenue
FROM orders_clean;
```

```
SELECT category, COUNT(*) as count
FROM orders_clean
GROUP BY category
ORDER BY count DESC;
```

Fixing Words:

```
UPDATE orders_clean
SET category = 'Accessories'
WHERE category = 'Accesories';

SELECT category, COUNT(*) as count
FROM orders_clean
GROUP BY category
ORDER BY count DESC;
```

– End Of SQL (Lite) Executions –